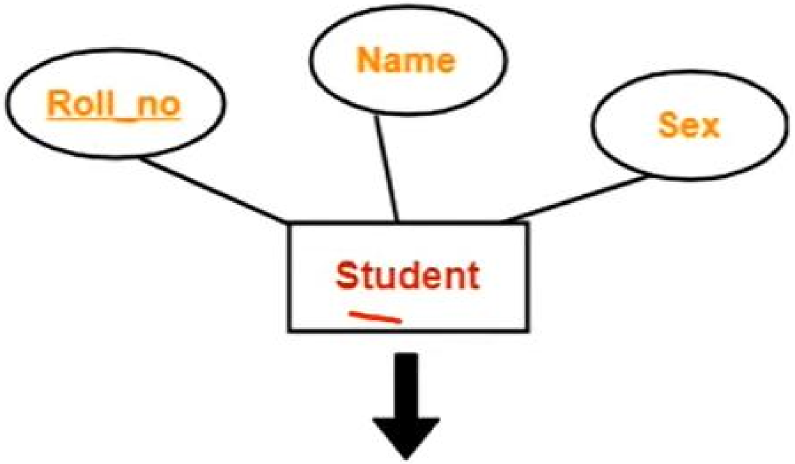
Converting ER Diagram into Table

Rule1: Strong Entity Set With Only Simple Attributes

- A strong entity set with only ***simple attributes*** will require only **one table** in relational model.
  - Attributes of the table will be the attributes of the entity set.
  - The primary key of the table will be the key attribute of the entity set.

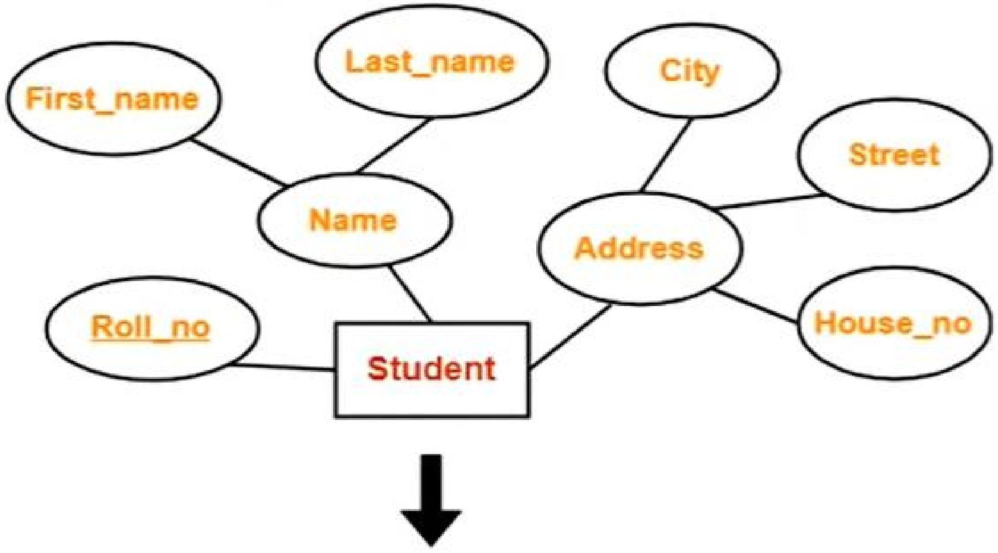


Roll_no	Name	Sex

Schema : Student (Roll_no , Name , Sex)

Rule 2: Strong Entity Set With Composite Attributes

- A strong entity set with any number of ***composite attributes*** will require **only one table** in relational model.
- While conversion, simple attributes of the composite attributes are taken into account and not the composite attribute itself.

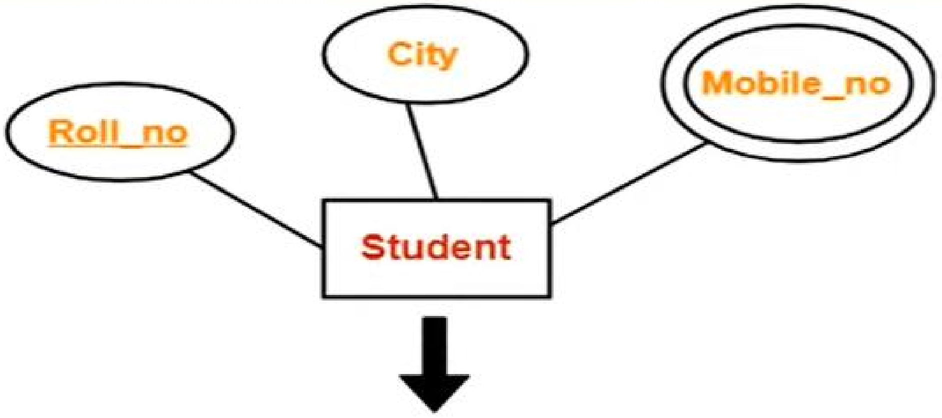


Roll_no	First_name	Last_name	House_no	Street	City

Schema : Student (Roll_no , First_name , Last_name , House_no , Street , Ci

Rule 3: Strong Entity Set With Multi Valued Attributes

- A strong entity set with any number of ***multi-valued attributes*** will require **two tables** in relational model.
  - One table will contain all the simple attributes with the primary key.
  - Other table will contain the primary key and all the multi valued attributes.



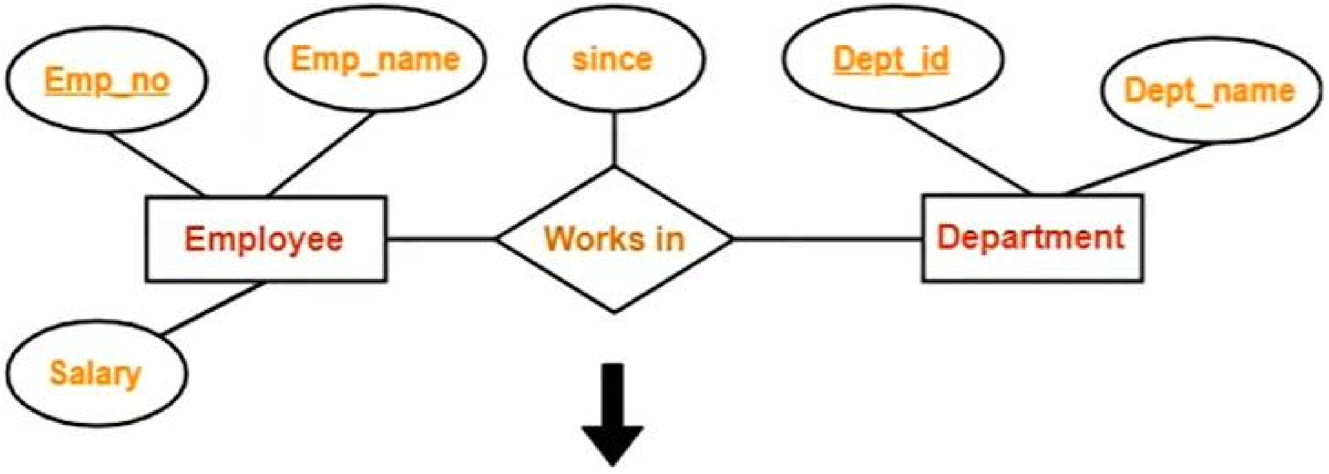
Roll_no	City

Roll_no	Mobile_no

Need Two tables in Multivalued Attribute

Rule 4: Translating Relationship Set into a Table

- A ***relationship set*** will require **one table** in the relational model.
- Attributes of the table are:
  - Primary key attributes of the participating entity sets
  - Its own descriptive attributes if any.
- Set of non-descriptive attributes will be the primary key
- For given ER diagram, three tables will be required in relational model-
  - One table for the entity set "Employee"
  - One table for the entity set "Department"
  - One table for the relationship set "Works in"



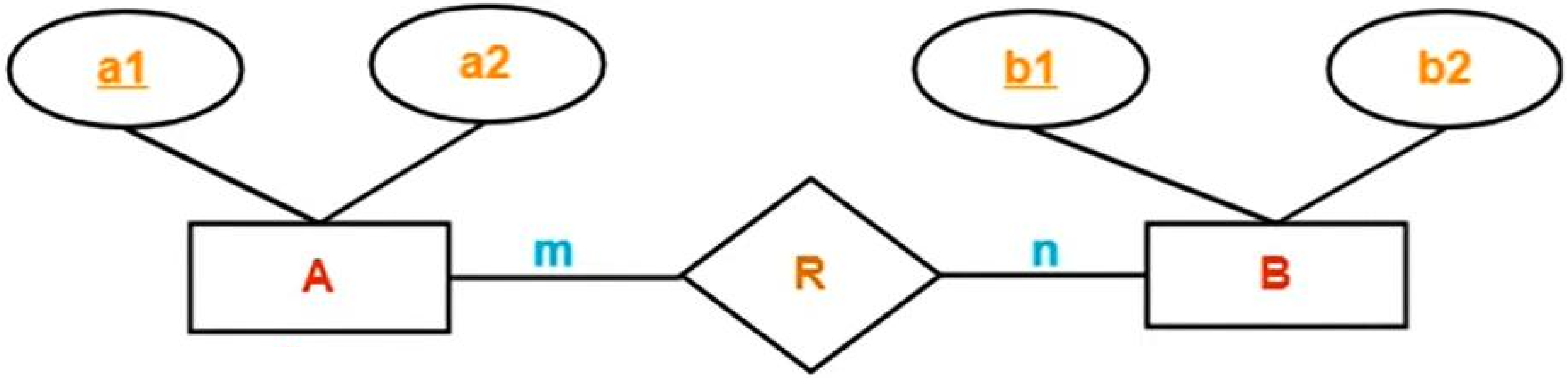
Emp_no	Dept_id	since

Schema : Works in (Emp_no , Dept_id , since)

Rule 5: For Binary Relationships With Cardinality Ratios

- The following four cases are possible-
  - **Case-1:** Binary relationship with cardinality ratio m:n
  - **Case-2:** Binary relationship with cardinality ratio 1:n
  - **Case-3:** Binary relationship with cardinality ratio m:1
  - **Case-4:** Binary relationship with cardinality ratio 1:1

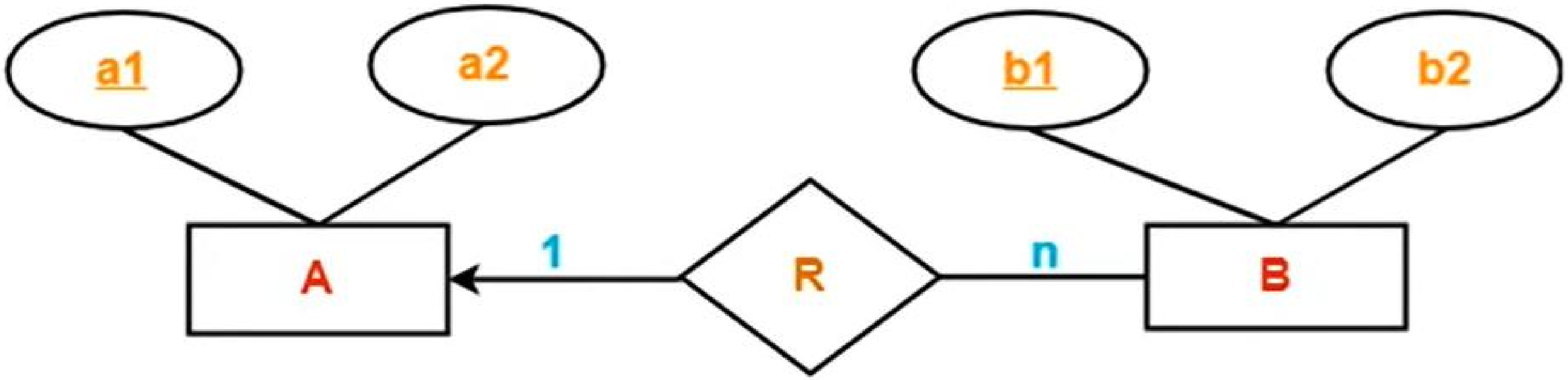
Case-1: For Binary Relationship with Cardinality Ratio m:n



In ***Many-to-Many relationship***, **three tables** will be required-

1. A (<u>a1</u> , a2)
2. R (<u>a1</u> , <u>b1</u>)
3. B (<u>b1</u> , b2)

Case-2: For Binary Relationship with Cardinality Ratio 1:n

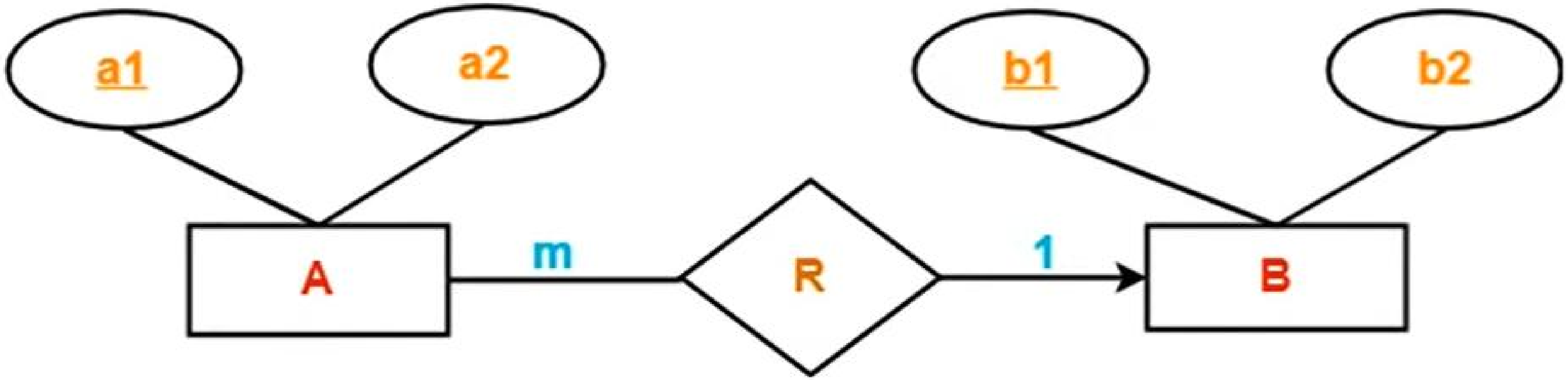


In ***One-to-Many relationship***, **two tables** will be required-

1. **A (<u>a1</u> , a2)**
2. **BR (<u>b1</u> , b2, a1)**

<u>NOTE</u> - Here, combined table will be drawn for the entity set B and relationship set R.

Case-3: For Binary Relationship with Cardinality Ratio m:1

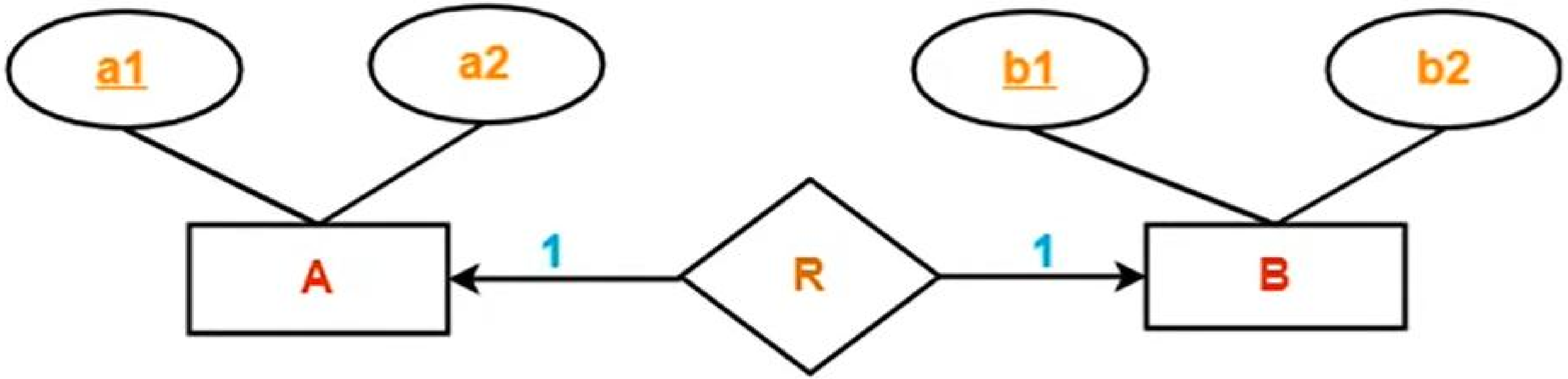


In ***Many-to-One relationship***, **two tables** will be required-

1. **AR (<u>a1</u> , a2 , b1)**
2. **B (<u>b1</u> , b2)**

<u>NOTE</u> - Here, combined table will be drawn for the entity set A and relationship set R.

Case-4: For Binary Relationship with Cardinality Ratio 1:1



In ***One-to-One relationship***, **two tables** will be required. Either combine 'R' with 'A' or 'B'

Way-01:

1. AR (<u>a1</u> , a2 , b1)
2. B (<u>b1</u> , b2)

Way-02:

1. A (<u>a1</u> , a2)
2. BR (<u>b1</u> , b2, a1)

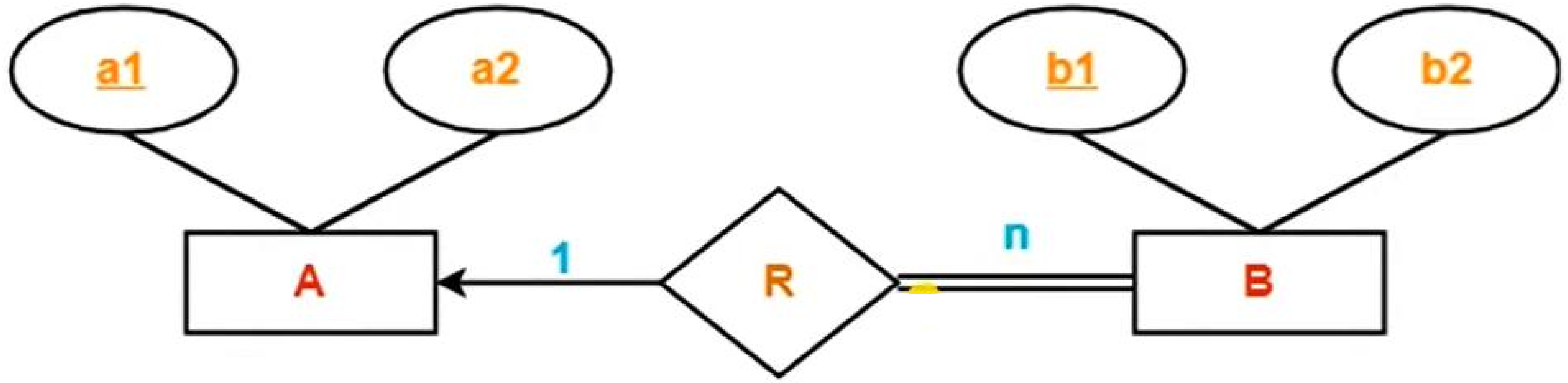
(For determining minimum number of tables)

- **While determining the minimum number of tables required for binary relationships with given cardinality ratios**, following thumb rules must be kept in mind-
  - For binary relationship with cardinality ration **m : n** ,
    - *Separate and individual tables* will be drawn for each entity set and relationship (i.e. **three tables** will be required).
  - For binary relationship with cardinality ratio either **m : 1 or 1 : n** ,
    - Always remember "many side will consume the relationship" i.e. a combined table will be drawn for many side entity set and relationship set (i.e. **Two tables** will be required).
  - For binary relationship with cardinality ratio **1 : 1** ,
    - **Two tables** will be required. You can combine the relationship set with any one of the entity sets.

Rule-6: For Binary Relationship with Both Cardinality Constraints and Participation Constraints

- Cardinality constraints will be implemented as discussed in Rule-5.
- Because of the *total participation constraint*, foreign key acquires **NOT NULL** constraint i.e. now foreign key can not be null.
- Two Cases:
  - **Case-1:** For Binary Relationship with Cardinality Constraint and Total Participation Constraint from One Side
  - **Case-2:** For Binary Relationship with Cardinality Constraint and Total Participation Constraint from Both Sides

Case-1: For Binary Relationship with Cardinality Constraint and Total Participation Constraint from One Side

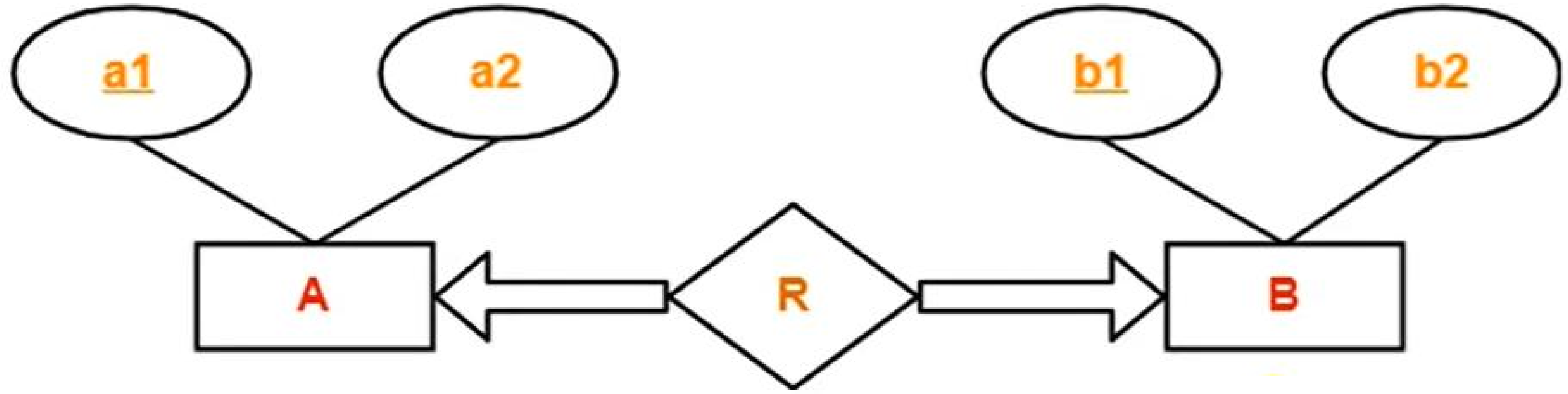


- Because cardinality ratio = 1 : n , we will combine the entity set B and relationship set R.
- Then, **two tables** will be required-
  1. A (<u>a1</u> , a2)
  2. BR (<u>b1</u> , b2 , a1)

Because of total participation, ***foreign key a1*** has acquired ***NOT NULL constraint***, so it can't be null now.

Case-2: For Binary Relationship with Cardinality Constraint and Total Participation Constraint from Both Sides

- If there is a key constraint from both the sides of an entity set with total participation, then that binary relationship is represented using **only single table.**

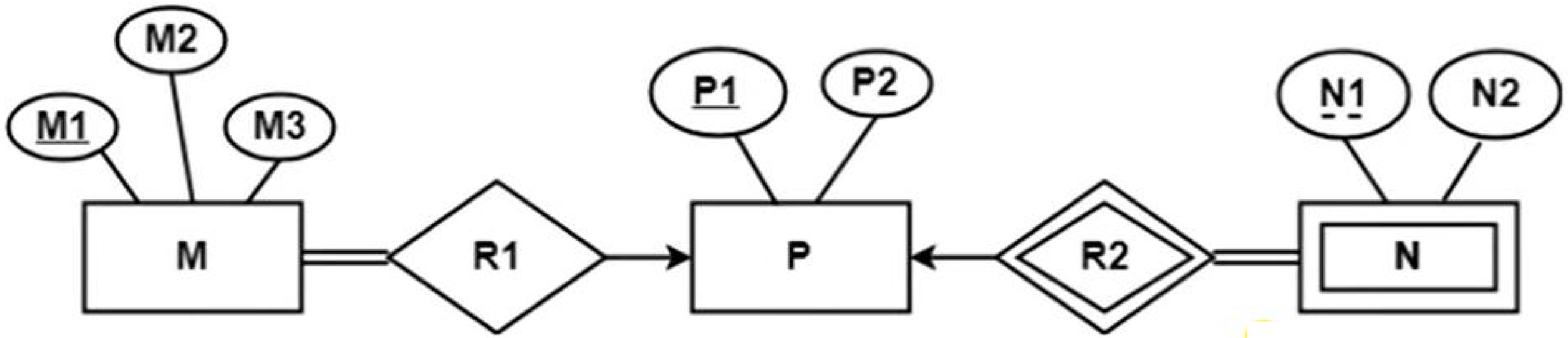


- Here, Only **one table** is required.

1. **ARB (<u>a1</u> , a2 , <u>b1</u> , b2)**

Question 1:

- Find the minimum number of tables required for the following ER diagram in relational model



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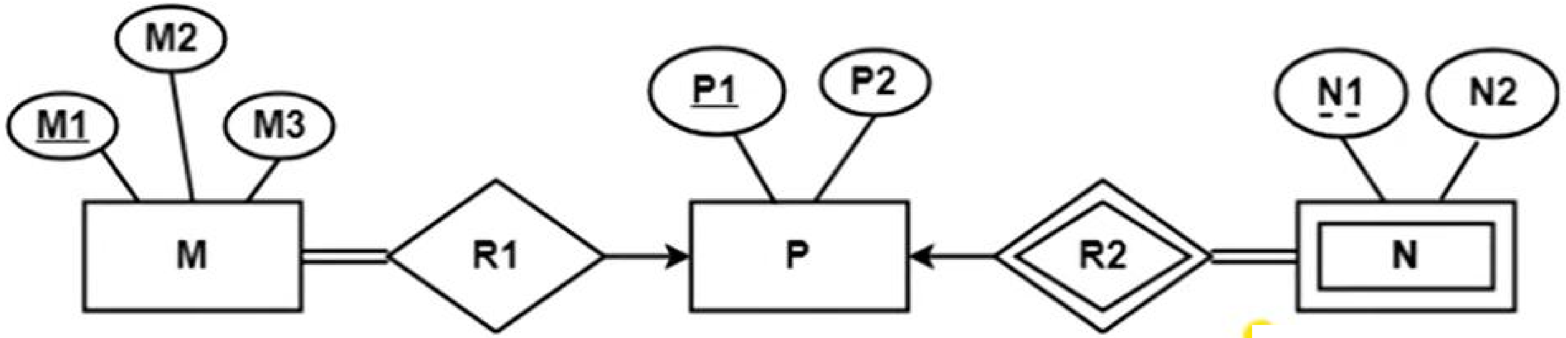

Solution-

Minimum 3 tables will be required-

1. MR1 (<u>M1</u> , M2 , M3 , P1)
2. P (<u>P1</u> , P2)
3. NR2 (<u>P1</u> , <u>N1</u> , N2)

Question 2:

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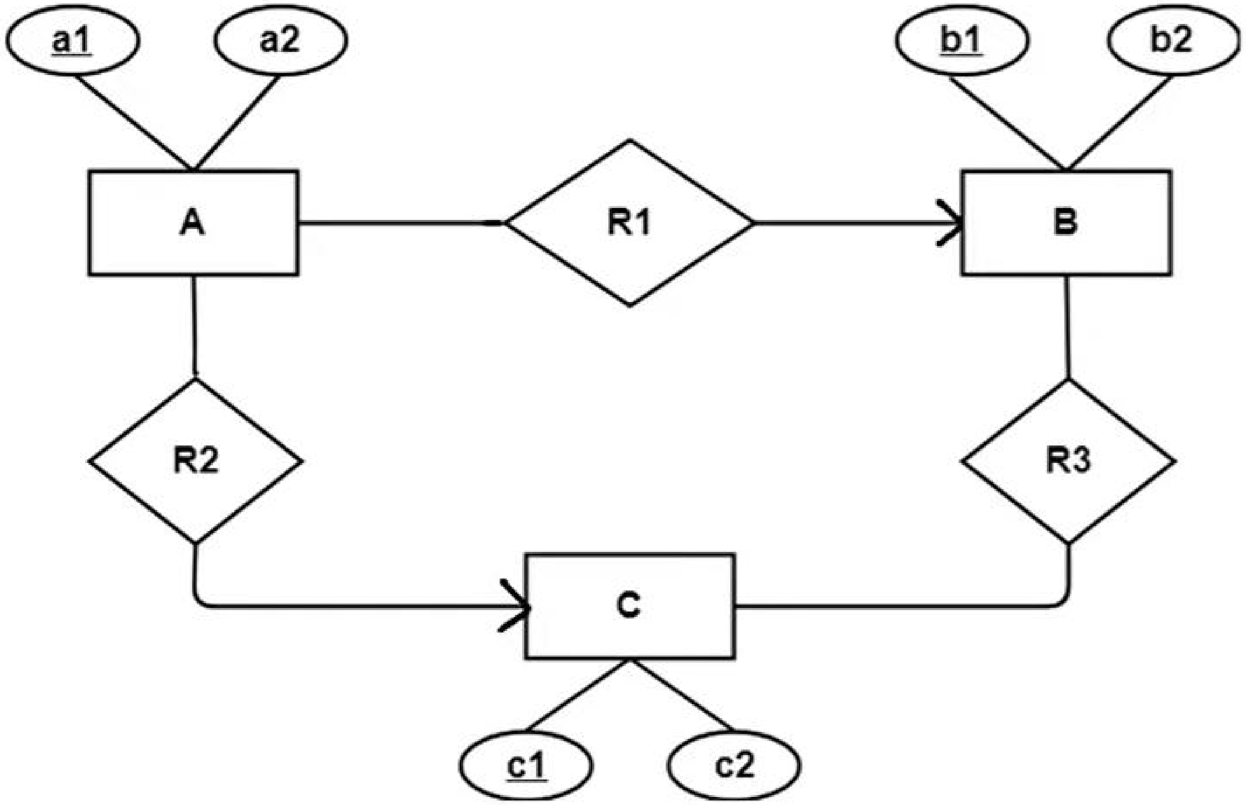

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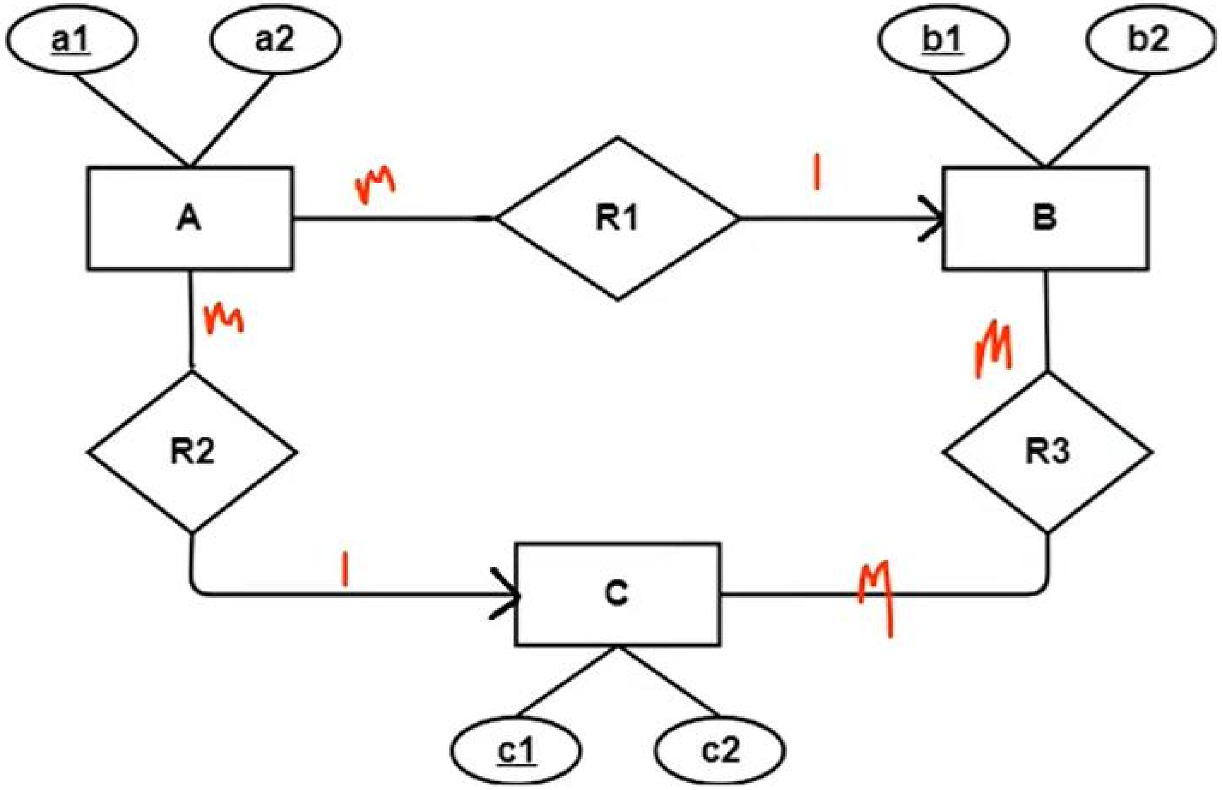


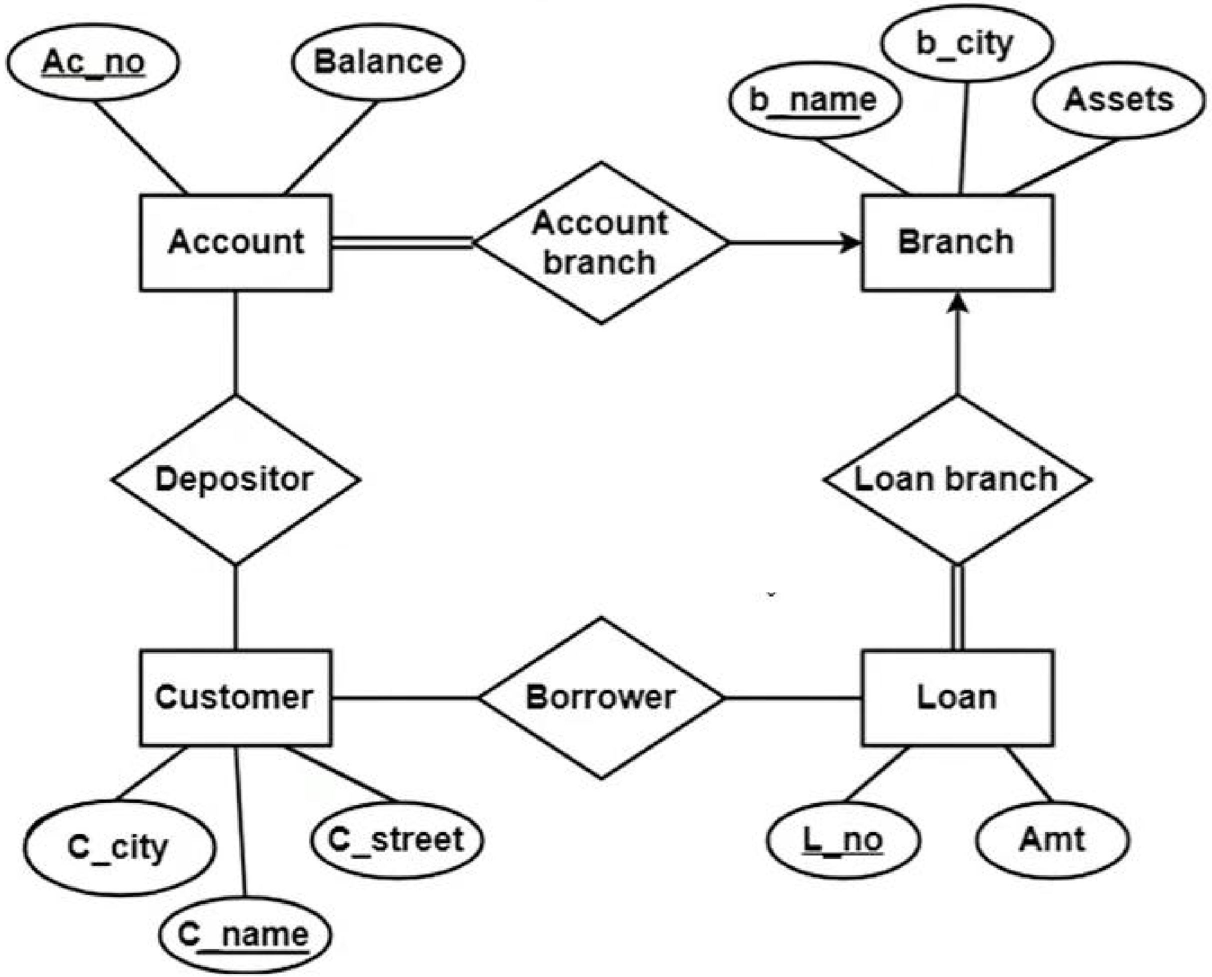
Solution-

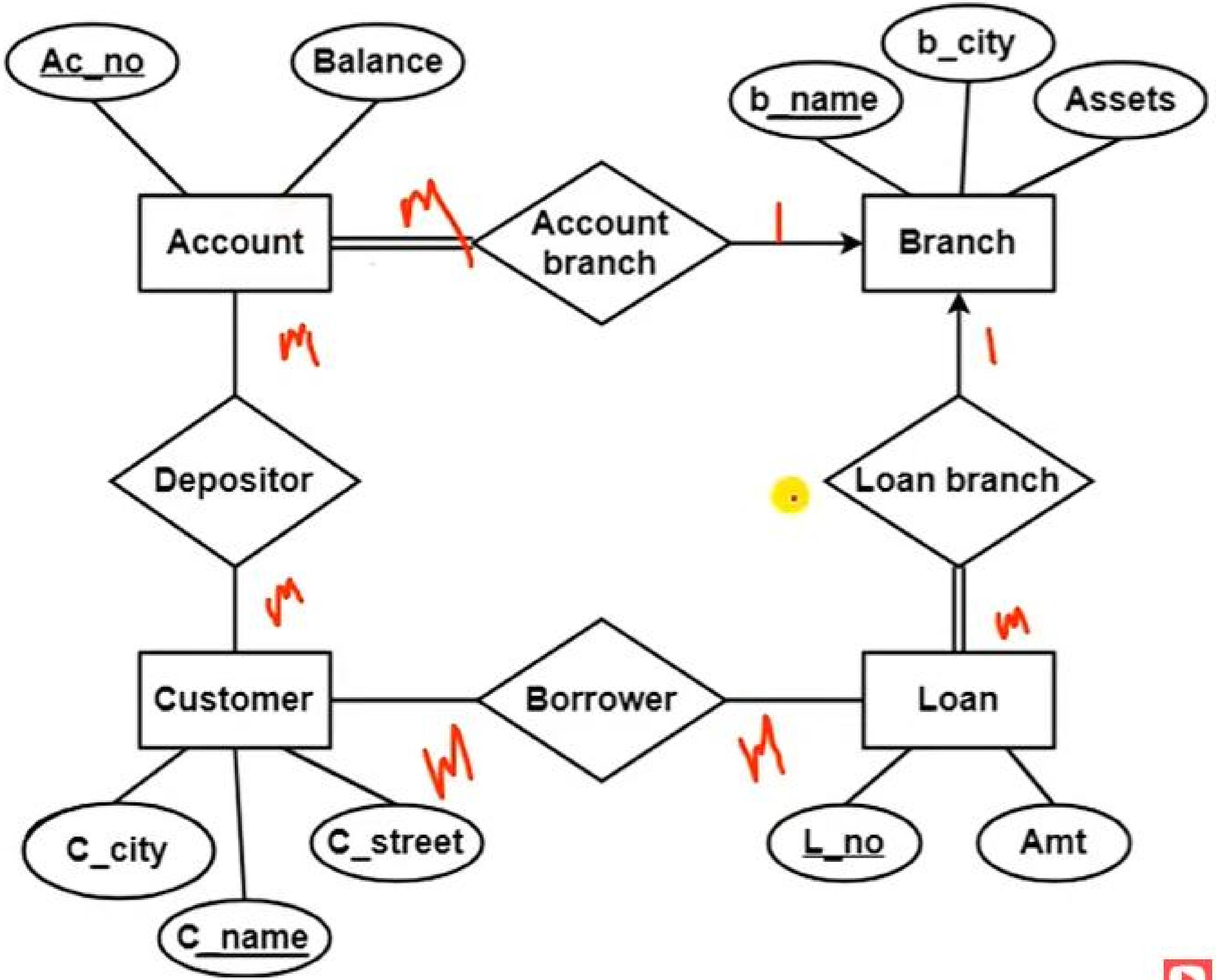
Minimum 4 tables will be required:

1. AR1R2 (<u>a1</u> , a2 , b1 , c1)
2. B (<u>b1</u> , b2)
3. C (<u>c1</u> , c2)
4. R3 (<u>b1</u> , <u>c1</u>)

Question 1:

- Find the minimum number of tables required for the following ER diagram in relational model




Ac_no
Balance
b_city
b_name
Assets
Account
Account branch
Branch
M
1
M
1
Depositor
Loan branch
M
M
Customer
Borrower
Loan
M
M
C_city
C_street
C_name
L_no
Amt

Question 1:

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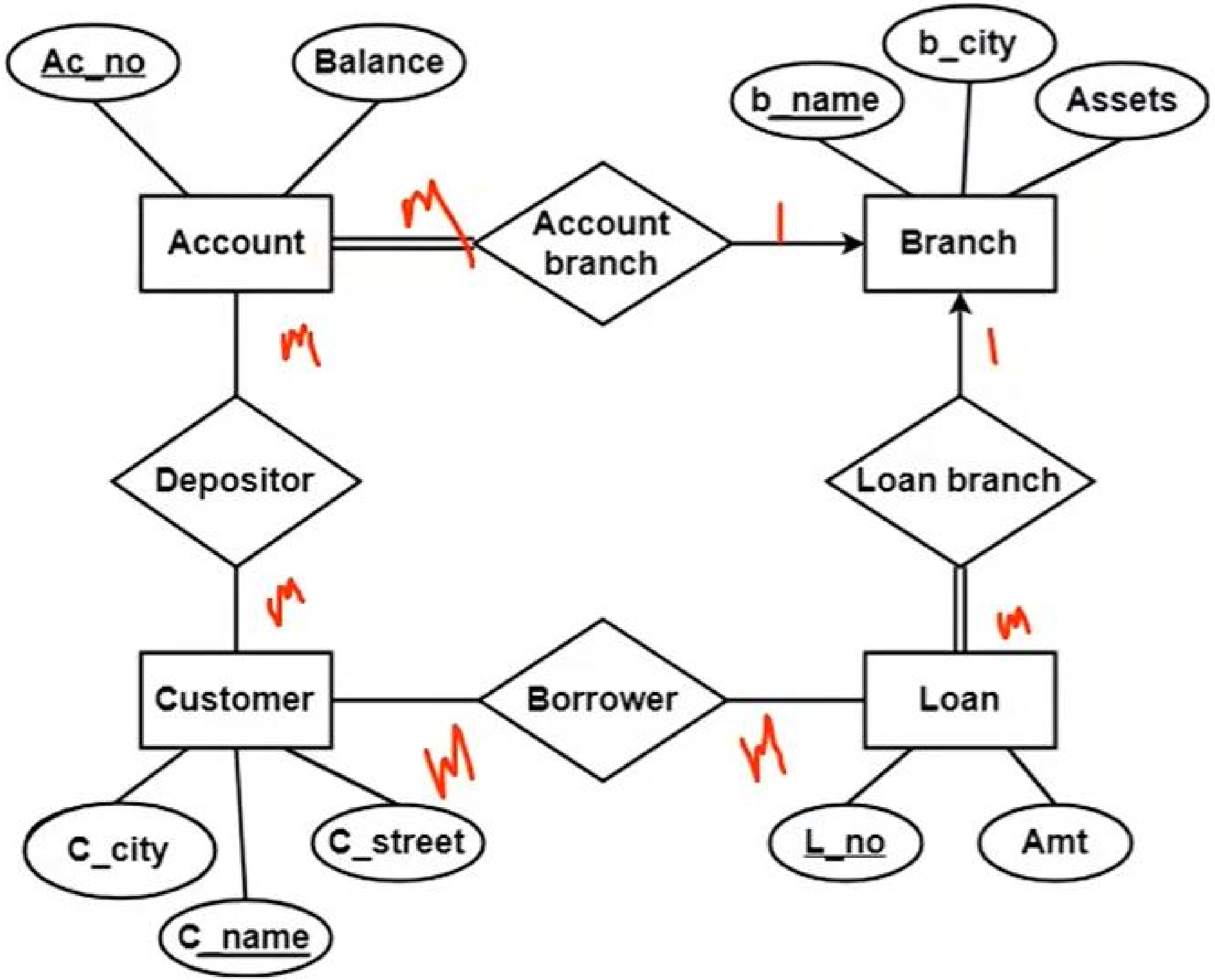


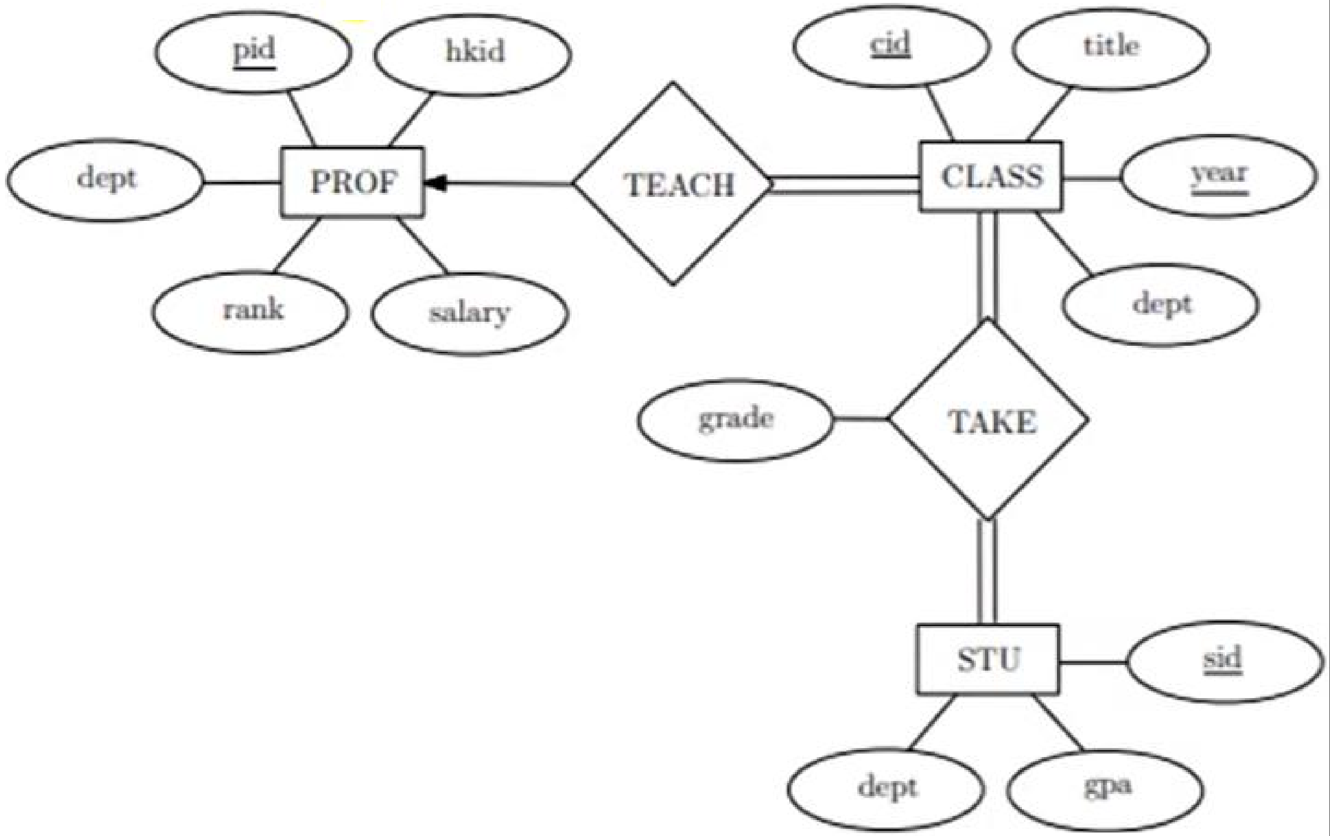
Solution-

Minimum 6 tables will be required:

1. Account (Ac_no , Balance , b_name)
2. Branch (b_name , b_city , Assets)
3. Loan (L_no , Amt , b_name)
4. Borrower (C_name , L_no)
5. Customer (C_name , C_street , C_city)
6. Depositor (C_name , Ac_no)

Question 2:

- Find the minimum number of tables required for the following ER diagram in relational model



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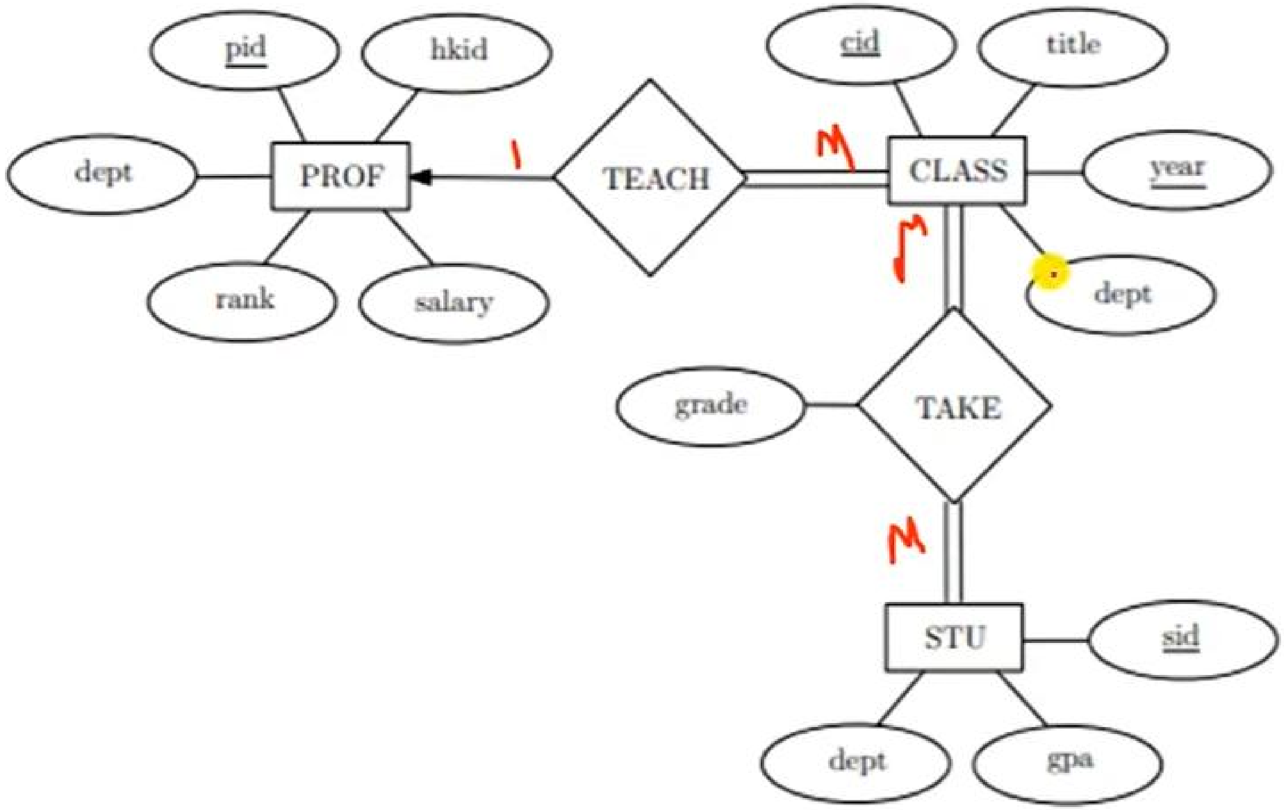
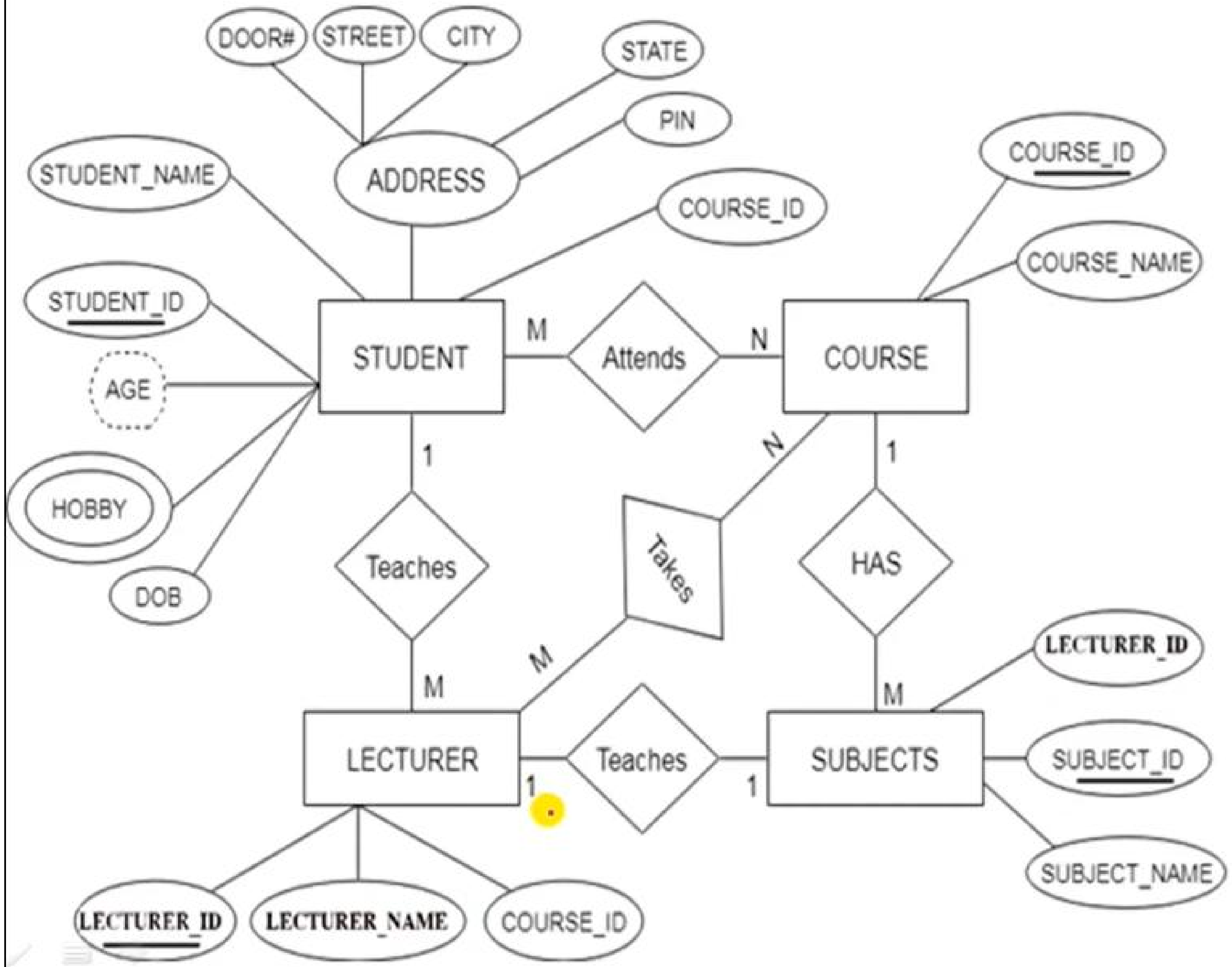


Solution-

Applying the rules, minimum 4 tables will be required:

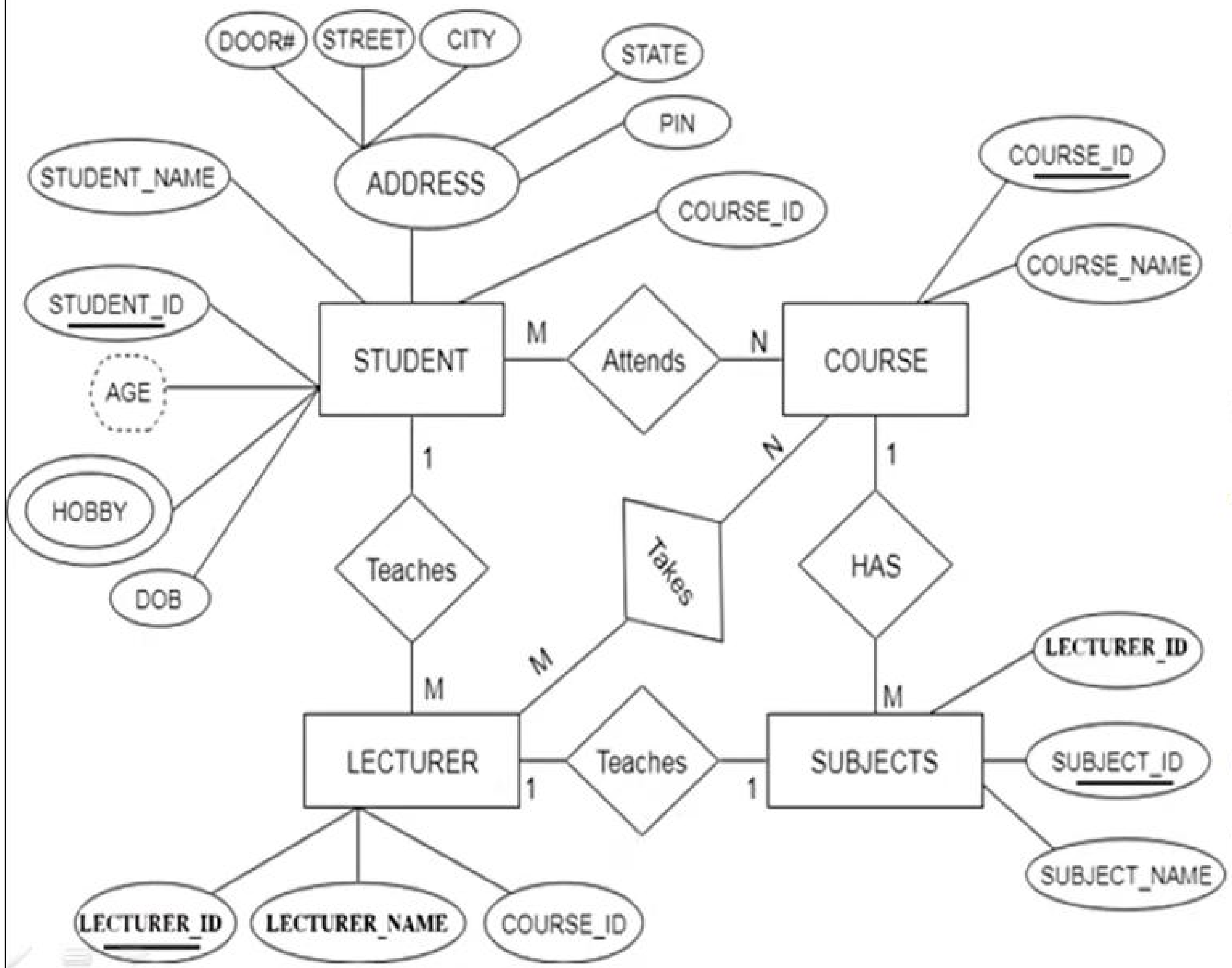
1. PROF (pid, hkid, dept, rank, salary)
2. CLASS (cid, year, title, dept, pid)
3. STU (sid, dept, gpa)
4. TAKE (cid, year, sid, grade)

Question 3:



DOOR#
STREET
CITY
STATE
PIN
ADDRESS
STUDENT_NAME
COURSE_ID
STUDENT_ID
AGE
HOBBY
DOB
STUDENT
M
Attends
N
COURSE
COURSE_ID
COURSE_NAME
1
Teaches
M
N
Takes
M
1
HAS
M
LECTURER
Teaches
SUBJECTS
1
1
LECTURER_ID
SUBJECT_ID
SUBJECT_NAME
LECTURER_ID
LECTURER_NAME
COURSE_ID

Question 3:



Solution:

Minimum 7 tables will be required:

1. **Student** (Student_Id, Student_Name, DOB, Door#, Street, City, State, Pin, Course_Id)
2. **Lecturer** (Lecturer_Id, Lecturer_Name, Course_Id, Student_Id, Subject_Id)
3. **Course** (Course_Id, Course_Name)
4. **Subjects** (Subject_Id, Subject_Name, Lecturer_Id, Course_Id)
5. **Hobby** (Subject_Id, Hobby)
6. **Attends** (Subject_Id, Course_Id)
7. **Takes** (Lecturer_Id, Course_Id)